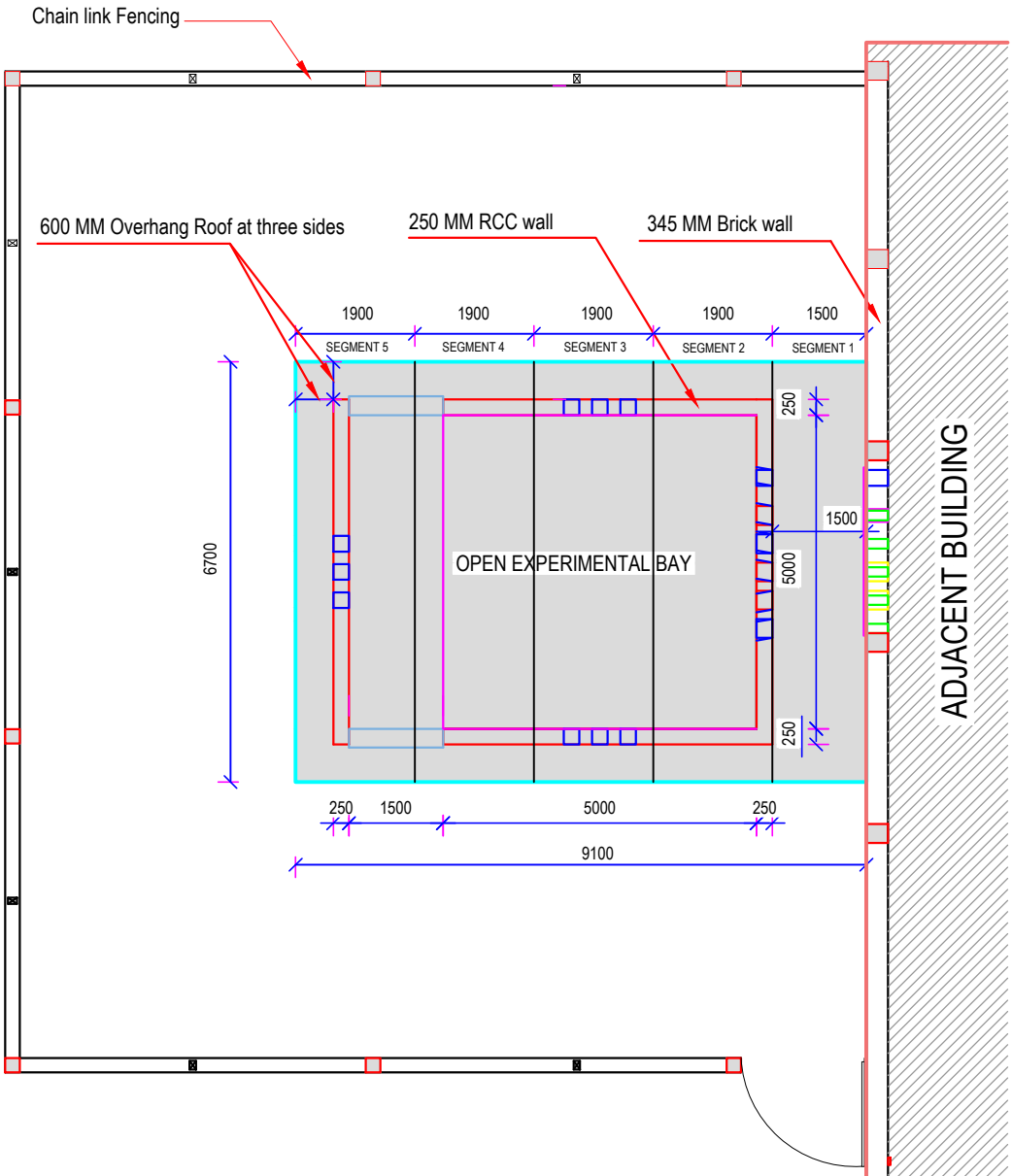
Technical Specifications: Motorized Retractable Roof for ADRL Experimental Bay

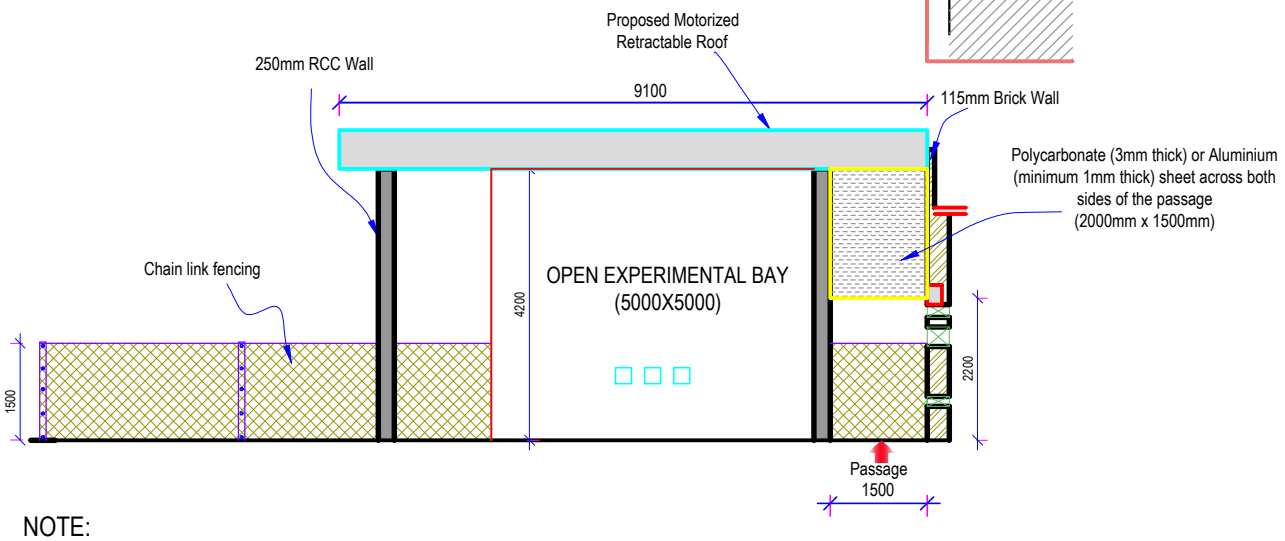
Indent Title: "Supply & installation of motorized retractable roof for ADRL Experimental Bay"

Sl. No	Specifications
1	Supply and installation of motorized retractable roof for ADRL experimental bay with the following specifications:
2.	**Size of Retractable roof to cover the Experimental Bay:** Experimental Bay Size: 8.5m x 5.5m x 4.2m (LxBxH) Roofing Area: 9.1m x 6.7m (including 0.6m overhang at three sides) Refer sketch for dimensions (Annexure 1)
3.	**Type of retractable roof:** Sliding/Telescopic Type with minimum 5 segments to cover the entire bay (including the fixed segment).
4.	**Material of construction: As per specifications 4.1 to 4.4**
4.1	**Roofing sheet:** • Non-corrosive Aluminium Alloy Panels (Powder Coated/Anodic Oxide Coated) with minimum thickness of roofing panel of 1 mm.
4.2	**Sealing of Enclosures:** Sealing of between Aluminum panels and structural members shall be provided (100% leak proof and 100% water resistant).
4.3	**Structural members:** • Track Mounting Base frame structure above the walls to suitably hold the retractable roof: Stainless Steel 304 grade • Tracks/Guides: Non-corrosive Aluminium Alloy (Powder Coated/Anodic Oxide Coated) extruded tracks • Profiles/Frames: Non-corrosive Aluminium Alloy (Powder Coated/Anodic Oxide Coated) hollow extruded profiles • Any additional steel structures, brackets and fasteners for the assemblies to hold the retractable roof: Stainless Steel 304 grade
4.4	• All the Aluminium structures shall be made with Non-corrosive Aluminium Alloy (Powder Coated/Anodic Oxide Coated). • All the steel structures shall be made with Stainless Steel 304 grade material.
5	**Mode of operation: As per specifications 5.1 to 5.5**
5.1	Mode: Provisions for Remote Operation as well as manual mode of operations shall be provided. Remote operation shall be from the adjacent building with approximate distance of 25 meters.
5.2	Remote Control Operated Gear Motor: • Rack and pinion arrangement • Power Supply: 230V AC, 50 Hz • Mechanical Limit Switch • SS304 Mounting Base • Built in Control Board
5.3	Motor shall be provided with proper sealing to have rain, humidity and dust protection. Motor shall be suitable for outdoor operations (IP 56 level is preferred).
5.4	Electrical Fittings: • End Limit Switches to indicate status LED on the control room 25 meters away from the site.

	• Control switches shall be provided near the site as well as inside the control room adjacent to site at a distance of 25 meters.
5.5	Safety interlocks for the electric motor shall be provided with interlock switches.
6	**General Terms: As per specifications 6.1 to 6.6**
6.1	**Warranty:** • Overall warranty of all the systems of the retractable roof should be minimum 2 years. • Warranty for individual components should be same as the warranty offered by the manufacturer for individual components.
6.2	**Annual Maintenance Contract (AMC):** • Party must quote a comprehensive AMC after the warranty period is over. • A separate order for AMC with detailed terms and conditions will be issued after the completion of the warranty period. • **Scope of AMC:** Visit based on Preventive Maintenance procedure and schedule any number of break down calls.
6.3	Transportation, unloading, scaffolding, installation charges including any additional stainless steel structures, electrical line routing from control room (25 meters away) and any other charges related to commissioning of the retractable roof shall be under the purview of the supplier.
6.4	The party must provide appropriate packing that should be sufficient to withstand, without limitation, rough handling during transit and exposure to extreme temperatures, salt and precipitation during transit and open storage.
6.5	Demonstration of operation as well as training to the operators shall be provided at site.
6.6	**Tendering:** • Mode of Tender: LT - Two part • Party must submit bid offer in 2 parts: (i) Technical & Commercial Bid (ii) Price Bid • The party may visit the site before submitting the offers for tender.
7	**Technical documents to be submitted by the vendor:**
7.1	• Detailed Compliance Statement against all the technical specifications • Schematic diagram describing the control systems • Detailed design drawing and 3D models of the structural components • Product Catalogue including catalogue of individual components • Standard Operating Procedure for the operation of the Retractable Roof in Remote Mode and Manual Mode • Warranty Certificate • User Manual

ANNEXURE 1





NOTE:

1. Size of Retractable Roof: Length - 9100mm, Width - 6700mm
2. Roofing area to be covered is shown as shaded region.
3. Overhang of 600mm must be present across three sides of the retractable roof.
4. Retractable Roof must have 5 segments with one fixed (Segment 1) & four movable segments (Segment 2, 3, 4, 5).
5. Additional structure of size 2000mm x 1500mm made of Polycarbonate (3mm thick) or Aluminium (minimum 1mm thick) sheet must be present across both sides of the passage.